



Explainable Vision Transformers for Histopathological Cancer Detection



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Deep Learning Course

Clinical Challenge: Cancer Detection in Histopathology



The Problem

Manual cancer detection is **time-consuming, subjective**, and requires specialized expertise



LC25000 Dataset

15,000 histopathology images • **768×768** resolution • 3 organ classes



Two Classification Tasks

Task 1: Colon

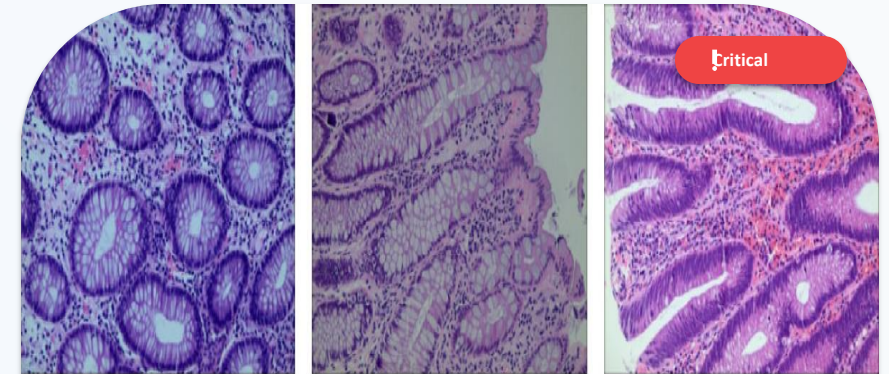
Binary Classification

Benign vs **Adenocarcinoma**

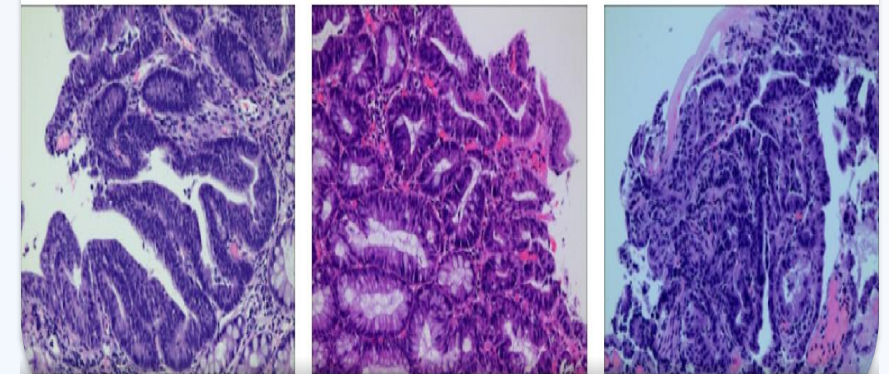
Task 2: Lung

3-Class Classification

Benign vs **Adenocarcinoma** vs **Squamous**



(a) Benign



Benign vs Malignant Tissue Structures

(b) Malignant



ViT-B/16: From Patches to Predictions



Patch Embedding

16×16 patches → 196 tokens + CLS token



Transformer Encoder

12 blocks • 768-dim embeddings • 12 attention heads



Classification Head

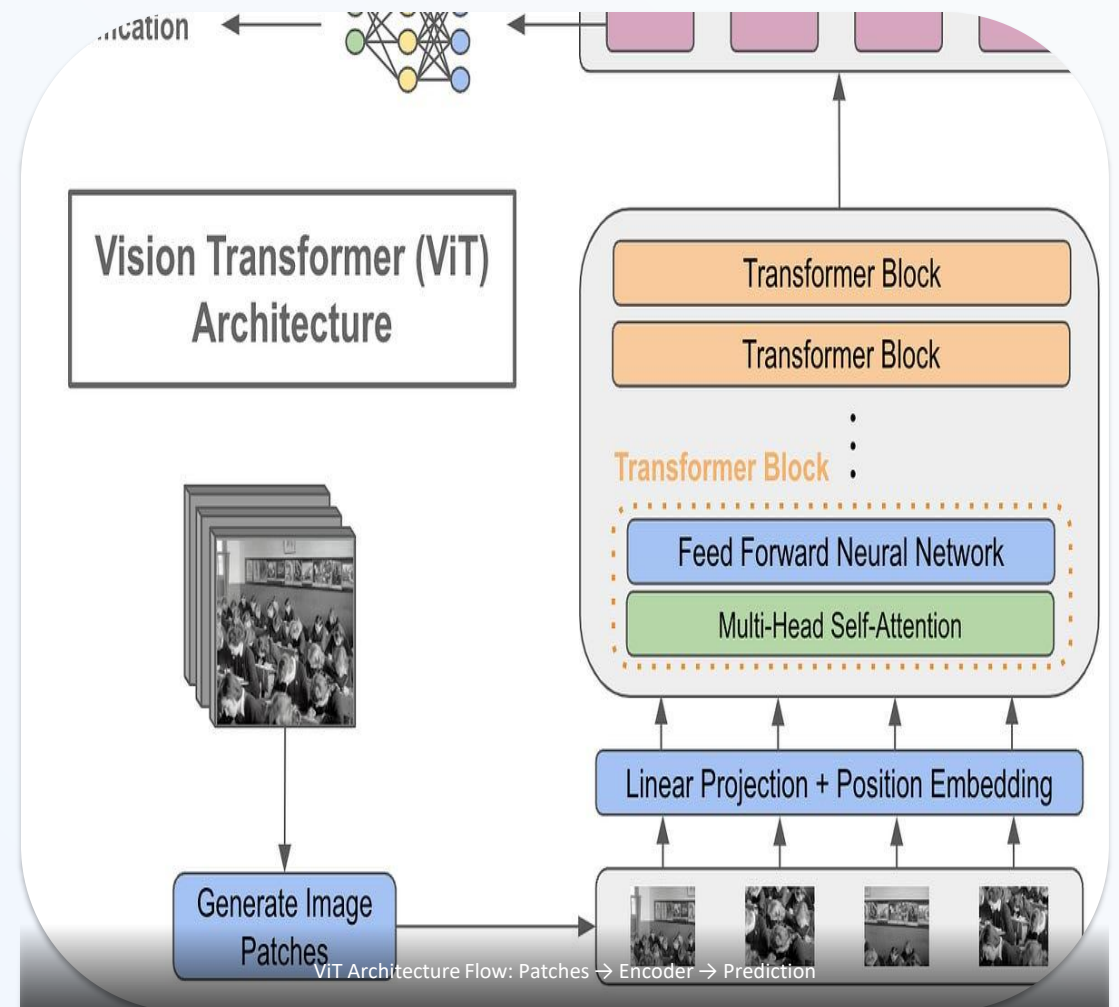
Linear(768→2) Colon

Linear(768→3) Lung



Transfer Learning

Pre-trained on ImageNet-21k • 86M parameters • Fine-tuned on LC25000



Optimization Pipeline for Medical Imaging



AdamW Optimizer

$lr=2e-4$ • $weight_decay=1e-4$ • Decoupled weight decay



Cosine Annealing LR Scheduler

Smooth decay • $T_max=15$ • Prevents abrupt convergence



Class-Weighted Cross-Entropy Loss

Handles class imbalance in lung 3-class task • Compensates minority classes



Aggressive Data Augmentation

RandomHorizontalFlip

RandomVerticalFlip

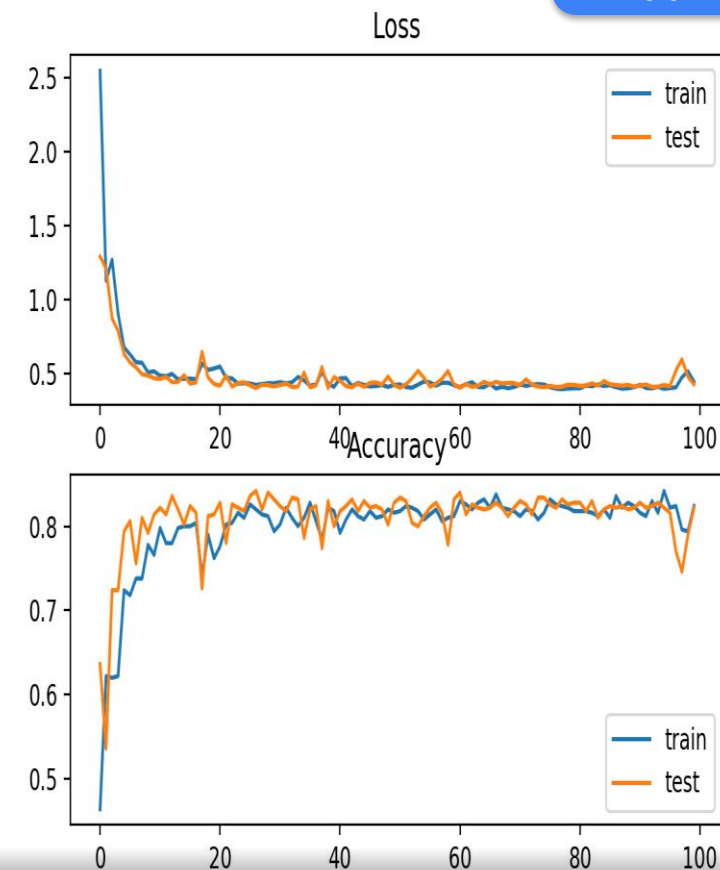
ColorJitter

RandomRotation(15°)



Best Model Checkpointing

Save model with highest validation accuracy



Cosine Annealing Schedule: Smooth LR Decay



Clinical-Grade Performance



Colon Classification

Test Accuracy

99.93%

Macro F1-Score

0.9993

Binary: Benign vs Adenocarcinoma



Lung Classification

Test Accuracy

99.82%

Macro F1-Score

0.9982

3-Class: Benign vs ACA vs SCC



Confusion matrix

True	Non-cancerous	6,074	605
	Cancerous	604	3,716
		Non-cancerous	Cancerous
		Predicted	

Colon: Binary Classification

Near-perfect diagonal



True Class

Predicted Class	Positive	Negative
Positive	True Positive	False Positive
Negative	False Negative	True Negative



Lung: 3-Class Classification

Robust across subtypes

Key Insight: Near-perfect discrimination between benign and malignant tissues across both organ sites

👁️ Transparent AI: From Black Box to Clinical Insight



Attention Rollout

Propagates attention weights through **12 transformer layers** to identify regions of interest



Grad-CAM for ViT

Class-discriminative visualization showing model focus for specific predictions



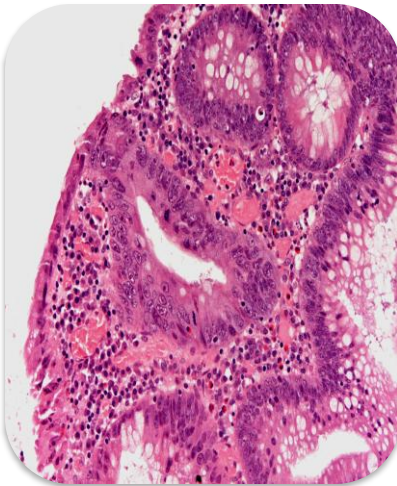
Dual XAI Approach

Provides both **global** (Attention Rollout) and **local** (Grad-CAM) explanations



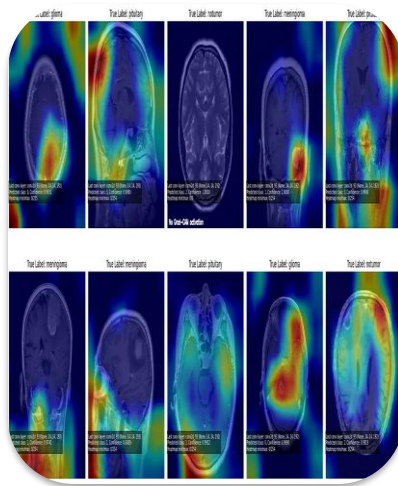
Clinical Significance

Heatmaps reveal model focuses on **histological structures**, not artifacts •
Critical for clinical trust and adoption



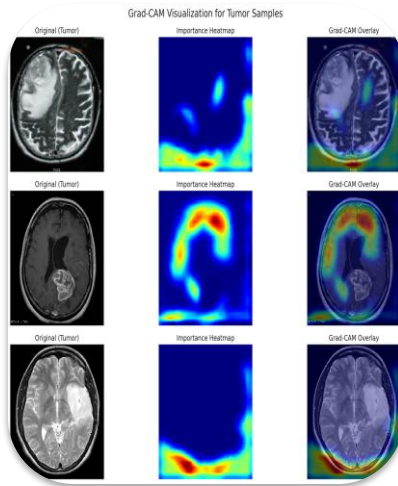
Original Image

Raw tissue sample



Attention Rollout

Global attention focus



Grad-CAM

Class-specific regions



Both methods agree on salient regions • Model focuses on **malignant tissue structures**



Bridging Deep Learning and Clinical Practice



Key Contributions

99.93% Colon accuracy • 99.82% Lung accuracy

Novel XAI for ViT in histopathology

Dual explanation builds clinical trust



Clinical Impact

- Explainability enables **pathologist verification**
- Reduces diagnostic errors & variability
- Foundation for **multi-organ** detection system



Future Directions

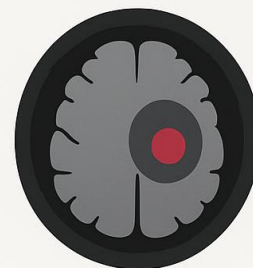
- **Multi-modal fusion** (histopathology + radiology + clinical)
- Extension to other cancer types
- Real-time clinical decision support



Understanding AI Vision: Grad-CAM vs Occlusion Sensitivity

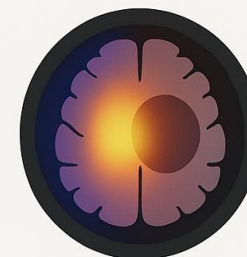
A comparison of model interpretation techniques in medical imaging

Original Brain MRI



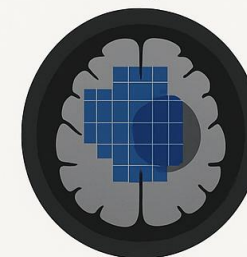
Input Image

Grad-CAM Visualization



- Gradient-based highlighting of important regions
- Fast computation (single pass)
- Internal network focus

Occlusion Sensitivity Analysis



- Systematically masking regions to measure impact
- Computationally intensive
- Higher precision

Comparing interpretation techniques for AI-powered brain tumor detection



ViT-Base (Ours)



CNN Baselines

→↵ Dual Explanation Framework: Attention Rollout vs Grad-CAM



Attention Rollout

Type: Class-agnostic

Layers: All 12 blocks

Filter: Discard ratio for noise reduction

Scope: Global focus

Fusion: Mean/Min/Max



Grad-CAM for ViT

Type: Class-discriminative

Source: Last block gradients

Focus: Prediction-specific features

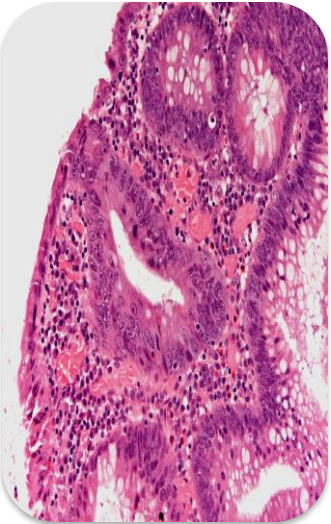
Scope: Local regions

Activation: Weighted sum + ReLU

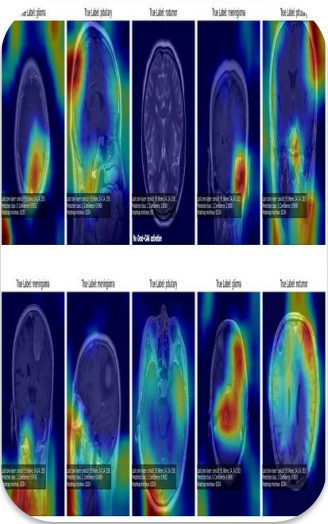


Complementary Insights

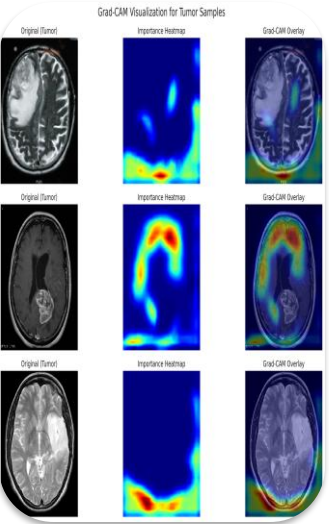
- **Rollout:** WHERE model looks overall
- **Grad-CAM:** WHAT drives each prediction
- **Together:** Complete explanation



Original Image
Raw tissue sample



Attention Rollout
Global attention flow



Grad-CAM
Class-specific focus



Dual approach confirms: Model focuses on **histological structures**, not artifacts • High confidence predictions correlate with precise attention



Future Direction: Multi-Modal Cancer Classification



Histopathology Branch

ViT-B/16 → **768-dim** features

High-resolution tissue analysis



Radiology Branch

ViT-S/16 → **384-dim** features

CT / X-Ray imaging data



Clinical Metadata Branch

MLP Encoder → **128-dim** features

Age, smoking, stage, biomarkers



Fusion Strategy

Concatenate: $768 + 384 + 128 = 1280$ dim

Fusion MLP → Classification



Key Advantage: Combines visual patterns with clinical context → **more robust, interpretable** predictions for clinical decision support



Scalable Framework: From Lung to Multi-Organ Detection



Organ-Agnostic Pipeline

Same ViT-B/16 backbone • Custom heads • Unified XAI



Easy Extension

1. Register new organ in config
2. Specify classes & data path
3. Auto-generate custom head



Key Benefits

- **Consistent XAI** across organs
- **Standardized metrics** & evaluation
- **Rapid deployment** for new types



Lung

3 Classes
Benign • ACA • SCC



Colon

2 Classes
Benign • Adenocarcinoma



Breast

2 Classes
Benign • Malignant



Input Data



ViT-B/16 Engine



XAI Layer



Diagnosis



Scalable Design: `build_model_for_organ('organ_name')` → Auto-configure ViT + custom head + XAI tools



Medical-Grade Data Augmentation for Robust Training



Why Augmentation Matters

- No canonical orientation in histopathology
- Stain variation between laboratories
- Limited training data → prevent overfitting



Train-Only Augmentations

RandomHorizontalFlip

Tissue rotation simulation

RandomVerticalFlip

Slide orientation simulation

RandomRotation(15°)

Placement variation

ColorJitter

Staining compensation

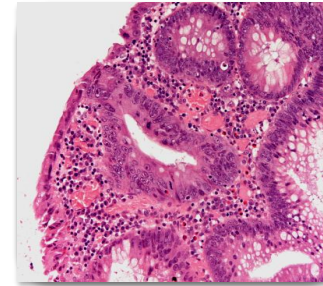
RandomResizedCrop

Different field of view simulation

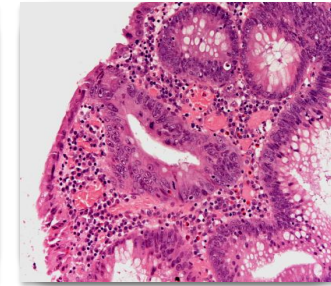


Evaluation Protocol

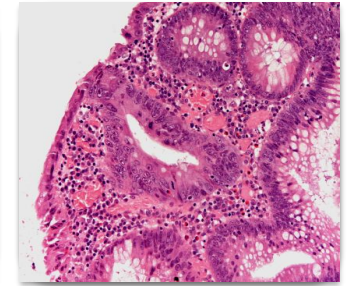
Clean transforms only: Resize + Normalize • No augmentation → Unbiased metrics



Original

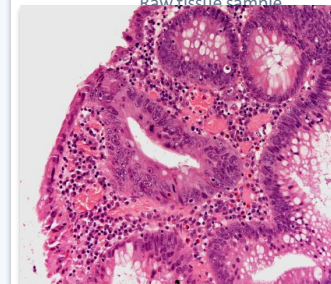


Horizontal Flip



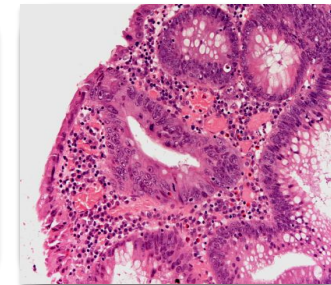
Color Jitter

Stain variation



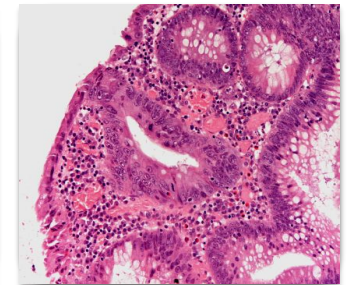
Rotation (15°)

Slide placement



RandomCrop

Field of view



Vertical Flip

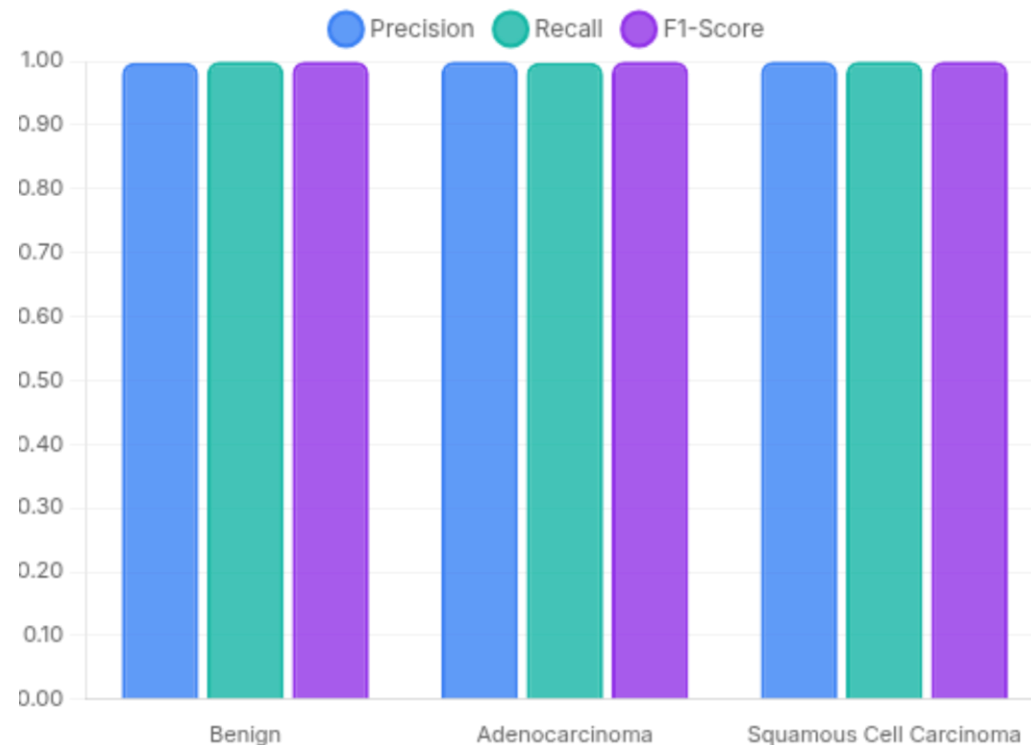
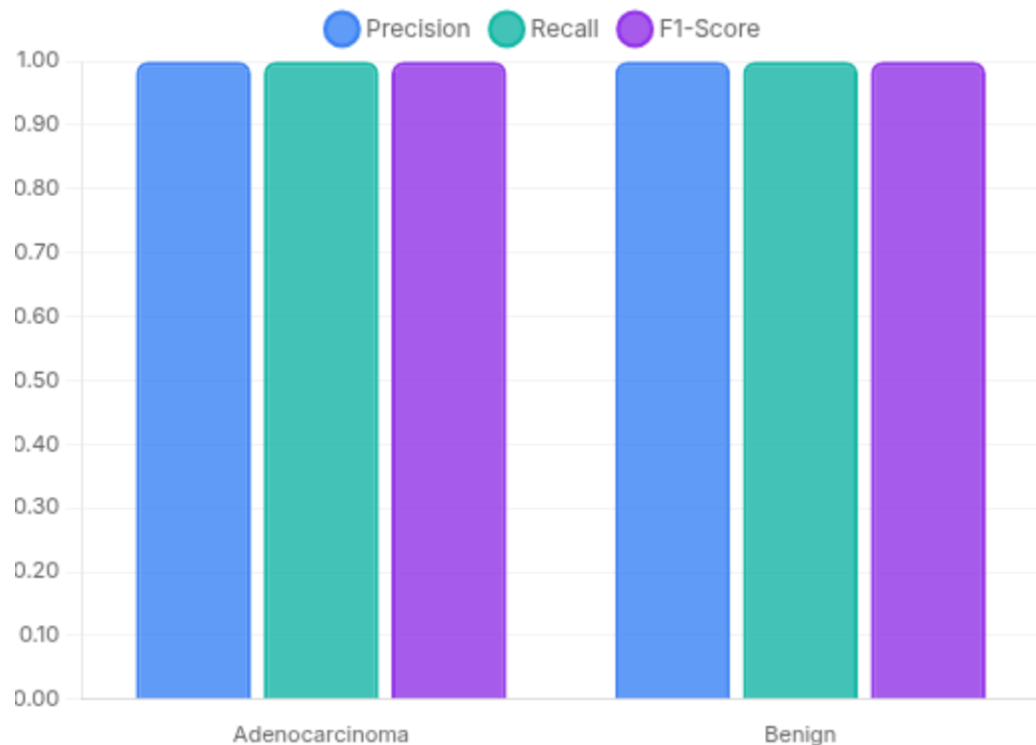
Orientation variation



Clinical Insight: Augmentation simulates **real-world variability** in tissue preparation and imaging → Improves generalization to unseen clinical samples



Robust Performance Across Cancer Subtypes



Key Insight: No significant performance gap between classes • Class-weighted loss ensures fairness • Macro F1 scores consistently exceed 0.998 across all subtypes